Forensic Verification of Marketplace Listings

Detecting AI-generated product images and verifying description claims with a single frozen vision backbone

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Deep Learning for Visual Analytics (DLVA) · Indian Institute of Science · Project Proposal

Topic of Choice

Fraud on second-hand marketplaces is moving from stolen stock photos to *AI-generated* product photos, often paired with text that promises more than the picture shows. I propose a buyer-side **listing verifier** that takes a photo and its description and returns two judgements:

- **Authenticity.** Is the photo a real capture, or was it produced by a diffusion model?
- **Consistency.** Do the checkable claims in the description (colour, item count; material as a stretch goal) agree with what is visible in the image?

Both judgements come from one frozen CLIP ViT-L/14 backbone with small trainable heads, covering three suggested themes: misinformation in manipulated images, vision + language, and robustness to real-world degradation.

High-Level Overview of the Topic

Fake-image detection has moved from CNNs trained on one generator's artefacts to reusing frozen foundation features; three recent papers shape this plan.

> **CVPR 2023** **Towards Universal Fake Image Detectors that Generalize Across Generative Models**
>
> Utkarsh Ojha, Yuheng Li, Yong Jae Lee
>
> A linear probe on *frozen* CLIP features beats end-to-end CNNs on unseen GANs and diffusion models: a feature space never trained to spot fakes does not learn the generator-specific shortcut.

> **ICLR 2025** **A Sanity Check for AI-generated Image Detection**
>
> Shilin Yan, Ouxiang Li, Jiayin Cai, Yanbin Hao, Xiaolong Jiang, Yao Hu, Weidi Xie
>
> Introduces *Chameleon*, generated images that fool human annotators; most detectors drop to near chance. Their AIDE model fuses DCT-frequency patches with CLIP semantics, showing that both cues are needed.

> **CVPR 2025** **Community Forensics: Using Thousands of Generators to Train Fake Image Detectors**
>
> Jeongsoo Park, Andrew Owens
>
> 2.7M images from more than 4,800 generators; generalisation improves with the *number of generators* seen in training more than with the number of images.

Real photos and ground-truth claims come from **ABO** (Collins et al., CVPR 2022): 147k Amazon listings with structured colour, material and item-count fields under CC BY 4.0, so no text parsing is needed. A small open vision–language model (SmolVLM-500M) answers the attribute questions.

Proposed Research Direction

Gaps in existing work

- Detectors are tested on scenes and faces, never on clean studio product photos; transfer to this domain is unreported.
- Images are judged alone. In a listing the fraud often lives in the *text–image relation*, which no detector checks.
- Robustness to JPEG, downscaling and screenshots, the normal path from seller to buyer, is reported only in passing.

Proposed approach

- **Fakes from ABO itself** via SDXL-Turbo (text- and image-to-image), so real and fake share framing and background; split by product family to stop leakage.
- **Authenticity:** CLIP linear probe + small head on log-DCT spectra, late fusion; released UnivFD and AIDE weights as baselines.
- **Consistency:** CLIP zero-shot contrast and multiple-choice VLM answers per claim, calibrated on synthetic mismatch pairs; 100 hand-labelled listings as a check.

Evaluation. AUROC and TPR at 5% FPR on seen vs. unseen generator families (a streamed CommunityForensics subset); AUROC against JPEG quality 90→30 for probe-only, spectrum-only and fused models; per-attribute F1 and calibration error for consistency. I expect colour and count to work and material to be the weak spot, and the frequency cue to collapse under compression while semantics hold; both are findings worth reporting.

References

[1] **Towards Universal Fake Image Detectors that Generalize Across Generative Models.** U. Ojha, Y. Li, Y. J. Lee. *CVPR*, 2023.

[2] **A Sanity Check for AI-generated Image Detection.** S. Yan, O. Li, J. Cai, Y. Hao, X. Jiang, Y. Hu, W. Xie. *ICLR*, 2025.

[3] **Community Forensics: Using Thousands of Generators to Train Fake Image Detectors.** J. Park, A. Owens. *CVPR*, 2025.

[4] **ABO: Dataset and Benchmarks for Real-World 3D Object Understanding.** J. Collins et al. *CVPR*, 2022.